



2011 Annual Examination

FORM V

MATHEMATICS EXTENSION 1

Wednesday 31st August 2011

General Instructions

- Writing time — 3 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.
- All necessary working should be shown in every question.
- Start each question on a new leaflet.

Structure of the paper

- Total marks — 135
- All nine questions may be attempted.
- All nine questions are of equal value.

Collection

- Write your name, class and master clearly on each leaflet.
- Hand in the nine questions in a single well-ordered pile.
- Hand in a leaflet for each question, even if it has not been attempted.
- If you use a second leaflet for a question, place it inside the first.
- Write your name on the question paper and place it inside your leaflet for Question One.

5A: BDD
5D: MK
5G: LJF

5B: PKH
5E: SJE
5H: SO

5C: FMW
5F: RCF
5I: MLS

Checklist

- Writing leaflets: 9 per boy.
- Candidature — 142 boys

Examiner
RCF

QUESTION ONE (15 marks) Start a new leaflet.

Marks

- | | |
|--|----------|
| (a) Write down the exact value of $\sin 315^\circ$. | 1 |
| (b) Express $\frac{1 - \sqrt{3}}{\sqrt{2}}$ with a rational denominator. | 1 |
| (c) Factorise $x^2 - 7x + 12$. | 1 |
| (d) Evaluate $\log_3 \frac{1}{9}$. | 1 |
| (e) Differentiate: | |
| (i) $x^5 + 2x$ | 1 |
| (ii) e^{2x+1} | 1 |
| (f) Write down a primitive of $x^3 - 2$. | 1 |
| (g) What is the nature of a stationary point if $\frac{d^2y}{dx^2} > 0$ at that point. | 1 |
| (h) Simplify $\ln(e^3)$. | 1 |
| (i) What is the natural domain of the function $f(x) = \ln(x + 2)$? | 1 |
| (j) Write as a single logarithm $2 \log_{10} a + \log_{10} 3b$. | 1 |
| (k) Sketch the graph of $(x - 2)^2 + y^2 = 4$, clearly indicating any intercepts with the axes. | 2 |
| (l) Solve $\tan \theta + \sqrt{3} = 0$, for $0^\circ \leq \theta \leq 360^\circ$. | 2 |

QUESTION TWO (15 marks) Start a new leaflet.

Marks

(a) (i) Sketch the parabola $y^2 = 4x$.

1

(ii) State the co-ordinates of the focus.

1

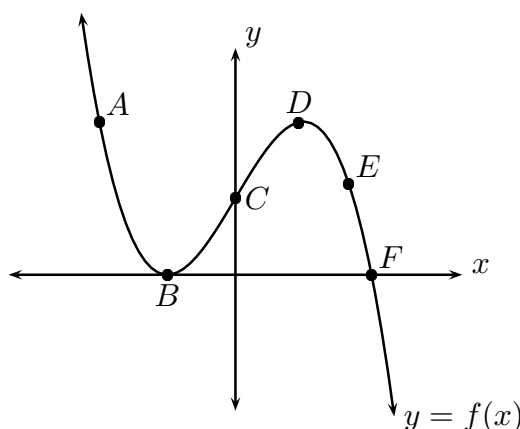
(b) Find the monic quadratic equation with roots 2 and -4 .

2

(c) The limiting sum of a GP with first term $\frac{1}{2}$ is 2. Find the common ratio.

2

(d)



Which point labelled in the diagram above has:

(i) $f'(x) < 0$ and $f''(x) > 0$,

1

(ii) $f''(x) = 0$.

1

(e) Given points $A(-2, 4)$ and $B(3, -11)$, find the co-ordinates of the point P which divides the interval AB internally in the ratio $2 : 3$.

2

(f) (i) Complete the square on the expression $x^2 + 6x - 1$.

2

(ii) Hence, or otherwise, find the co-ordinates of the vertex of the parabola $y = x^2 + 6x - 1$.

1

(g) (i) Graph the function $f(x) = \sqrt{4 - x^2}$.

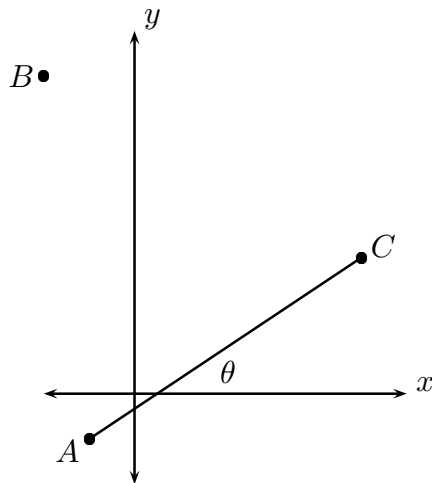
1

(ii) Hence evaluate the definite integral $\int_{-2}^2 \sqrt{4 - x^2} dx$ without using calculus.

1

QUESTION THREE (15 marks) Start a new leaflet.**Marks**

(a)



The points A , B and C , shown in the diagram above, have co-ordinates $(-1,-1)$, $(-2,7)$ and $(5,3)$ respectively. The angle between AC and the x -axis is θ .

- | | |
|--|---|
| (i) Find the gradient of AC . | 1 |
| (ii) Calculate the size of angle θ to the nearest degree. | 1 |
| (iii) Find the equation of line AC . Give your answer in general form. | 1 |
| (iv) Find the co-ordinates of D , the midpoint of AC . | 1 |
| (v) Show that AC is perpendicular to BD . | 1 |
| (vi) Explain why $\triangle ABC$ is isosceles. | 1 |
| (vii) Find the area of $\triangle ABC$. | 2 |
| (viii) Write down the co-ordinates of point E such that $ABCE$ is a rhombus. | 1 |
|
(b) | |
| (i) Differentiate $y = \log_e(2x - 1)$. | 1 |
| (ii) Hence find the gradient of the tangent to the curve $y = \log_e(2x - 1)$ at the point where $x = 3$. | 1 |
|
(c) | |
| (i) Graph $y = 2x - 2 $. Use at least one third of a page in your answer booklet for the graph. | 2 |
| (ii) Hence, or otherwise, solve the inequation $ 2x - 2 \geq 4$. | 2 |

QUESTION FOUR (15 marks) Start a new leaflet.

Marks

(a) Differentiate:

(i) $y = (5 - 2x)^7$ **2**

(ii) $y = \frac{x^4 + 2x^2}{x}$ **2**

(b) Find:

(i) $\int x\sqrt{x} \, dx$ **2**

(ii) $\int \frac{1}{2x^3} \, dx$ **2**

(iii) $\int \frac{1}{x-3} \, dx$ **1**

(iv) $\int e^{4x-2} \, dx$ **1**

(c) Evaluate $\int_{-1}^3 x(x-2) \, dx$. **2**

(d) Shade the region where the inequalities $x \geq 3$, $y \leq 10$ and $2x - y > 0$ are simultaneously satisfied. Clearly indicate on your diagram the inclusion or exclusion of points on the boundary lines and corners. **3**

QUESTION FIVE (15 marks) Start a new leaflet.**Marks**

(a) Solve $\sin(\alpha + 45^\circ) = \frac{1}{2}$, for $0^\circ \leq \alpha \leq 360^\circ$.

3

(b) The second term of a geometric series is 16 and the fifth term is 2.

(i) Find the common ratio.

1

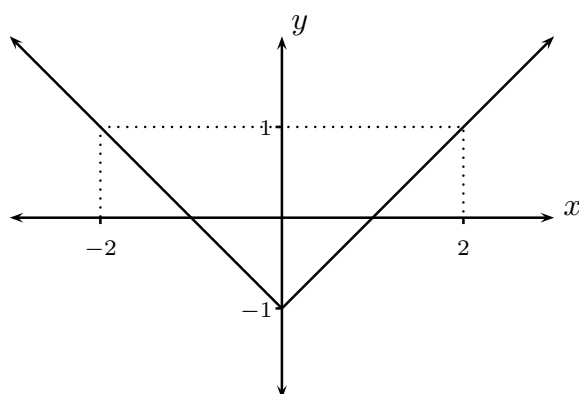
(ii) Find the first term.

1

(iii) Find the sum of the first fifteen terms. Write your answer as a rational number in simplest form.

2

(c)

1

The diagram above shows the graph of $y = |x| - 1$. Find the value of $\int_0^2 |x| - 1 \, dx$.

(d) Solve $\frac{2x - 6}{x} \leq 1$

3(e) Consider the parabola with equation $(y - 4)^2 = -16(x + 2)$.

(i) What is the focal length?

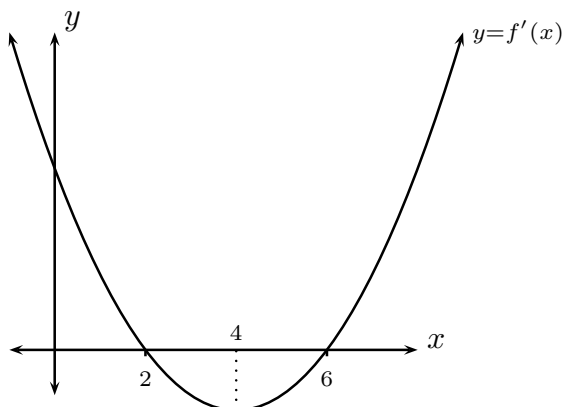
1

(ii) Sketch the curve, clearly showing the focus, the directrix and any intercepts with the axes.

3

QUESTION SIX (15 marks) Start a new leaflet.**Marks****3**

(a)



The diagram above shows the graph of the gradient function $y = f'(x)$ of the curve $y = f(x)$. Given that $f(0) = 0$, sketch the curve $y = f(x)$.

(b) Let $f(x) = \ln(x^2 + 1)$.

(i) Use the trapezoidal rule with three function values to estimate $\int_0^8 f(x) dx$. Write your answer correct to one decimal place. **2**

(ii) Use Simpson's rule with five function values to estimate $\int_0^8 f(x) dx$. Write your answer correct to three decimal places. **2**

(c) Differentiate:

(i) $y = \frac{\ln x}{x^2}$ **2**

(ii) $y = e^{2x}(2 - x^2)$ **2**

(d) Evaluate $\int_{-1}^0 \frac{4x}{x^2 + 3} dx$. **2**

(e) Solve the equation $9^x - 10 \times 3^x + 9 = 0$. **2**

QUESTION SEVEN (15 marks) Start a new leaflet.**Marks**

- (a) Suppose that the equation $2x^2 + 4x - 1 = 0$ has roots α and β . Without solving the equation, find the values of:

(i) $\alpha\beta$

1

(ii) $\frac{1}{\alpha} + \frac{1}{\beta}$

1

(iii) $(\alpha - \beta)^2$

2

(iv) $|\alpha - \beta|$

1

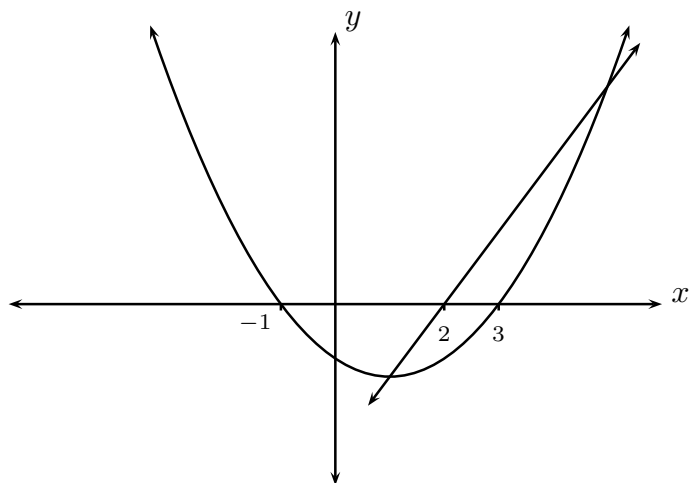
- (b) (i) By forming a suitable arithmetic sequence, find how many multiples of 7 lie between 100 and 1000?

2

- (ii) What is the sum of these multiples?

1

(c)



Consider the diagram above showing the parabola $y = 2x^2 - 4x - 6$ and the straight line $y = 8x - 16$.

- (i) Find the x co-ordinates of the points of intersection of the parabola and the line.

1

- (ii) Find the area enclosed between the parabola and the line.

3

- (d) (i) Show that the x co-ordinates of the points of intersection of the line $y = mx$ and the circle $(x + 3)^2 + y^2 = 1$ satisfy the equation $(1 + m^2)x^2 + 6x + 8 = 0$.

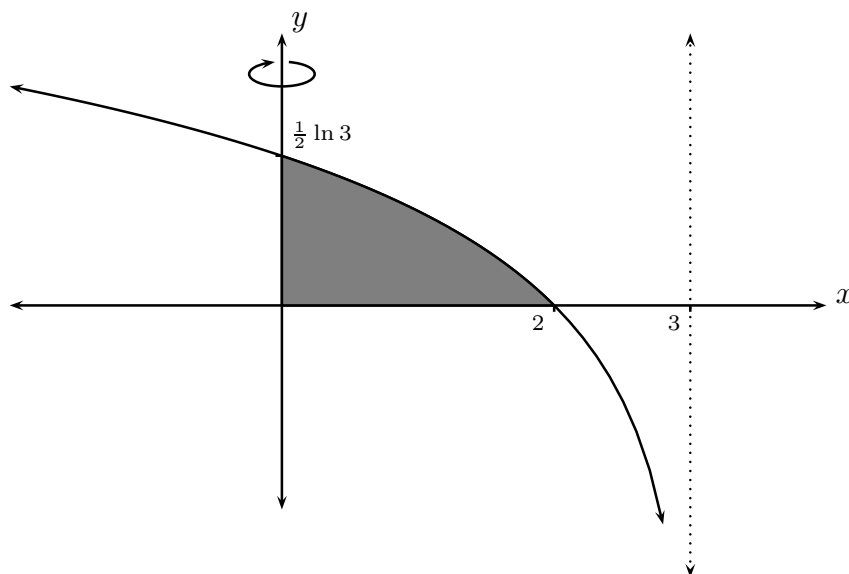
1

- (ii) Hence find the equations of the two tangents to the circle $(x + 3)^2 + y^2 = 1$ that pass through the origin.

2

QUESTION EIGHT (15 marks) Start a new leaflet.**Marks**

(a)



In the diagram above the shaded region is bounded by the curve $y = \frac{1}{2} \ln(3 - x)$ and the co-ordinate axes.

(i) Find x as a function of y .

1

(ii) Hence find the volume of the solid of revolution formed when the shaded region is rotated about the y -axis.

3

(b) Consider the curve $y = (x - 2)e^{-x}$.

(i) Find any x or y intercepts.

2

(ii) Find any stationary points and determine their nature.

3

(iii) Find any points of inflection.

3

(iv) By considering the behaviour of the function for large values of x , determine the equation of the horizontal asymptote.

1

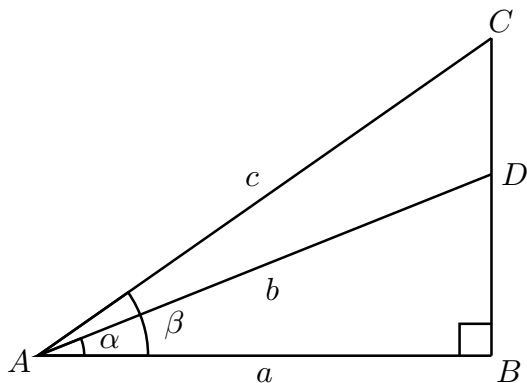
(v) Sketch the curve $y = (x - 2)e^{-x}$, clearly showing all the above information.

2

QUESTION NINE (15 marks) Start a new leaflet.

Marks

(a)



In the diagram above, $\angle BAD = \alpha$, $\angle BAC = \beta$ and sides AB , AD and AC have lengths a , b and c respectively.

(i) Write down an expression for the area of $\triangle ACD$ in terms of b , c , α and β .

1

(ii) Hence show that $\sin(\beta - \alpha) = \sin \beta \cos \alpha - \sin \alpha \cos \beta$.

2

(b) (i) Differentiate $\sqrt{x}e^{\sqrt{x}}$.

1

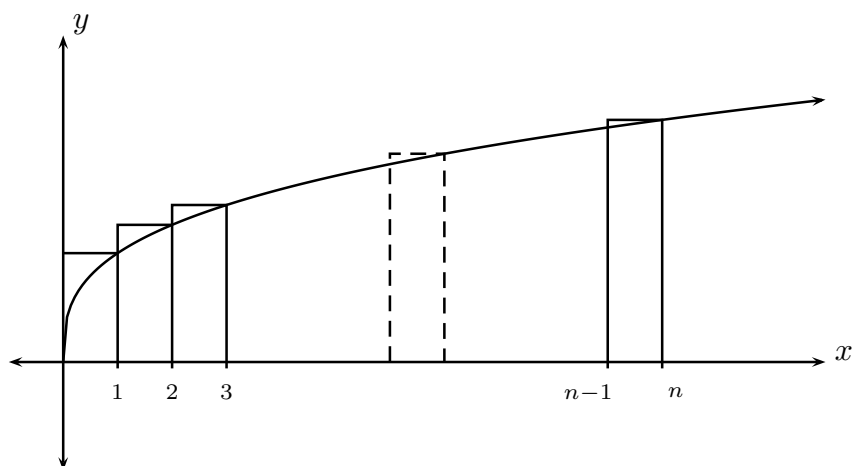
(ii) Hence find $\int e^{\sqrt{x}} dx$.

2

(iii) Evaluate $\int_0^4 e^{\sqrt{x}} dx$.

1

(c)



The diagram above shows the curve $y = \sqrt[3]{x}$, together with a set of n rectangles of unit width.

(i) By considering the areas of these rectangles, explain why

1

$$\sqrt[3]{1} + \sqrt[3]{2} + \sqrt[3]{3} + \dots + \sqrt[3]{n} > \int_0^n \sqrt[3]{x} dx.$$

(ii) By drawing another set of rectangles and considering their areas, show that

2

$$\sqrt[3]{1} + \sqrt[3]{2} + \sqrt[3]{3} + \dots + \sqrt[3]{n} < \int_1^{n+1} \sqrt[3]{x} dx.$$

(iii) Using parts (i) and (ii) find the value of $\sum_{n=1}^{100} \sqrt[3]{n}$, giving your answer accurate to as many significant figures as can be justified.

2

(d) Given that e^x is the limiting sum of the infinite series

3

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + \dots$$

where $n! = n \times (n-1) \times (n-2) \times \dots \times 2 \times 1$,

find an expression for the limiting sum of the infinite series

$$1 + 2x + \frac{3x^2}{2!} + \frac{4x^3}{3!} + \dots + \frac{(n+1)x^n}{n!} + \dots$$

END OF EXAMINATION

B L A N K P A G E

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

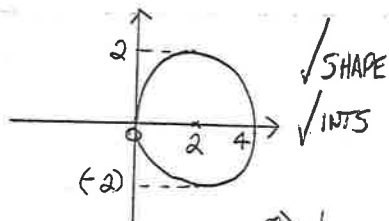
NOTE : $\ln x = \log_e x, \quad x > 0$

FIFTH FORM ANNUAL 2011

Question 1

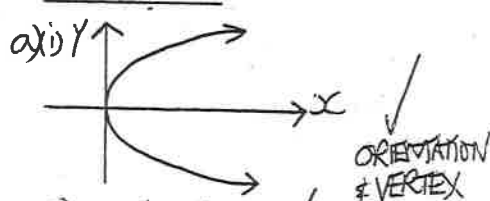
- a) $\sin 315^\circ = (-\sin 45^\circ)$ ✓
 $= (-\frac{1}{\sqrt{2}})$ ✓
b) $\frac{1-\sqrt{3}}{\sqrt{2}} = \frac{\sqrt{2}-\sqrt{6}}{2}$ ✓
c) $x^2-7x+12 = (x-3)(x-4)$ ✓
d) $\log_3 \frac{1}{9} = (-2)$ ✓
e) (i) $y = x^5 + 2x$
 $\frac{dy}{dx} = 5x^4 + 2$ ✓
(ii) $y = e^{2x+1}$
 $\frac{dy}{dx} = 2e^{2x+1}$ ✓
f) $f(x) = x^3 - 2$
 $F(x) = \frac{x^4}{4} - 2x + C$ ✓
g) Minimum turning point ✓
h) $\ln e^3 = 3$ ✓
i) $x+2 > 0$
 $x > (-2)$ ✓
j) $2\log_{10} a + \log_{10} 3b$
 $= \log_{10} (a^2 \times 3b) = \log_{10} 3a^2b$ ✓

k)



b) $\tan \theta = (-\sqrt{3})$
 $\theta = 120^\circ, 300^\circ$ ✓

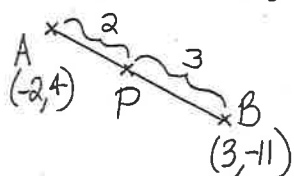
Question 2



- (ii) $S(1, 0)$ ✓
b) $(x-2)(x+4) = 0$ ✓ LHS
OR $x^2 + 2x - 8 = 0$ ✓ RHS
c) $S_\infty = \frac{a}{1-r}$ $2 = \frac{1}{1-r}$ ✓
 $1-r = \frac{1}{4}$
 $r = \frac{3}{4}$ ✓

- d) (i) Point A ✓
(ii) Point C ✓

e) $A(-2, 4)$ $B(3, -11)$ $2:3$

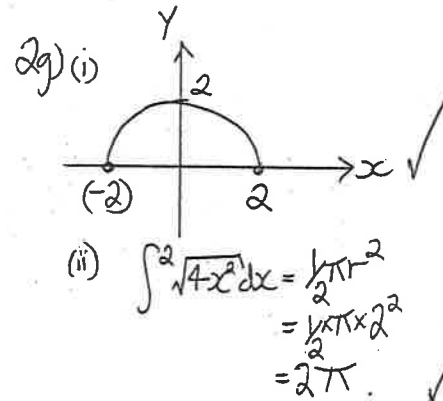


$x = \frac{2 \times 3 + 3 \times (-2)}{2+3}$ ✓
 $= 0$

$y = \frac{2 \times (-11) + 3 \times 4}{2+3}$ ✓
 $= (-2)$

P is $(0, -2)$ ✓

f) $x^2 + 6x - 1 = (x+3)^2 - 10$ ✓
(ii) $V(-3, -10)$ ✓



Question 3

- a) $DA(-1, -1)$ $C(5, 3)$
 $m_{DC} = \frac{3-(-1)}{5-(-1)} = \frac{4}{6} = \frac{2}{3}$ ✓
(ii) $\tan \theta = \frac{2}{3}$ $\theta = \tan^{-1} \frac{2}{3}$
 $\approx 34^\circ$ ✓
(iii) $y+1 = \frac{2}{3}(x+1)$
 $3y+3 = 2x+2$
 $0 = 2x-3y-1$ ✓
(iv) $D(\frac{-1+5}{2}, \frac{-1+3}{2}) = (2, 1)$ ✓
(v) $m_{BD} = \frac{7-1}{-2-2} = \frac{6}{-4} = -\frac{3}{2}$ show
 $\therefore m_{AC} \times m_{BD} = \frac{2}{3} \times (-\frac{3}{2}) = (-1)$ ✓
 $\therefore AC \perp BD$
(vi) $\triangle ABD \cong \triangle CBD$ (SAS) ✓
hence $AB = CB$ (Corresponding sides in congruent triangles)
OR
 $\triangle ABC$ is Isosceles since median BD is also altitude
OR
SHOW $AB = BC = \sqrt{65}$

(vii) Area = $\frac{1}{2} \times AC \times BD$
 $= \frac{1}{2} \times \sqrt{4^2 + 6^2} \times \sqrt{4^2 + 6^2}$ ✓
 $= \frac{1}{2} \times 52$
 $= 26$ ✓

(viii) $B \rightarrow A \xrightarrow{+1} -8$
hence $C \rightarrow E \xrightarrow{+1} -8$

$E(6, -5)$ ✓

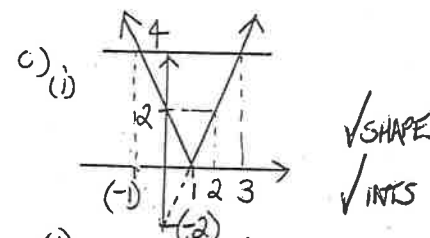
OR
Use D as midpoint of BE since diagonals of a rhombus bisect.

b) $y = \log_e(2x-1)$

$\frac{dy}{dx} = \frac{1}{2x-1} \times 2$
 $= \frac{2}{2x-1}$ ✓

(ii) $(\frac{dy}{dx})_{x=3} = \frac{2}{6-1} = \frac{2}{5}$ ✓

Gradient of tangent at $x=3$ is $\frac{2}{5}$



(ii) $x \leq (-1)$ OR $x \geq 3$ ✓

Question 4

a) i) $y = (5-2x)^7$
 $\frac{dy}{dx} = 7(5-2x)^6 \times (-2)$
 $= -14(5-2x)^6$ ✓✓

(ii) $y = \frac{x^4 + 2x^2}{x}$
 $= x^3 + 2x$ ✓
 $\frac{dy}{dx} = 3x^2 + 2$ ✓

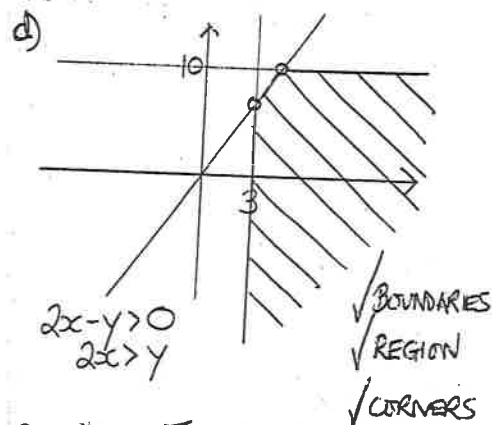
b) i) $\int x^{\frac{3}{2}} dx = \frac{x^{\frac{5}{2}}}{\frac{5}{2}} + C$
 $= \frac{2x^{\frac{5}{2}}}{5} + C$ ✓

(ii) $\int \frac{1}{2} x^{-3} dx = \frac{1}{2} \frac{x^{-2}}{-2} + C$
 $= -\frac{1}{4x^2} + C$ ✓

(iii) $\int \frac{1}{x-3} dx = \ln|x-3| + C$ ✓

(iv) $\int e^{4x-2} dx = \frac{e^{4x-2}}{4} + C$ ✓

c) $\int_{-1}^3 x(x-2) dx = \int_{-1}^3 x^2 - 2x dx$
 $= \left[\frac{x^3}{3} - x^2 \right]_{-1}^3$ ✓
 $= (9-9) - \left(-\frac{1}{3}-1\right)$ ✓
 $= \frac{4}{3}$ ✓



Question 5

a) $\sin(\alpha + 45^\circ)$ $0 \leq \alpha \leq 360^\circ$
 $45^\circ \leq \alpha + 45^\circ \leq 405^\circ$
 $\alpha + 45^\circ = 150^\circ, 390^\circ$ ✓
 $\alpha = 105^\circ, 345^\circ$ ✓

b) i) GP $t_2 = ar = 16$ ①
 $t_5 = ar^4 = 2$ ②
 $\frac{②}{①} \Rightarrow r^3 = \frac{2}{16} = \frac{1}{8}$
 $r = \frac{1}{2}$ ✓

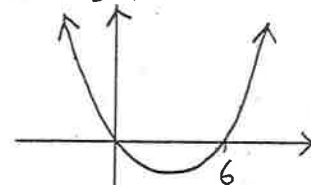
(ii) $ar = 16$
 $a = 16$
 $a = 32$ $t = 32$ ✓

(iii) $S = a \frac{1-r^{15}}{1-r}$
 $= 32 \frac{1-(\frac{1}{2})^{15}}{1-\frac{1}{2}}$ ✓
 $= 64 \left(1 - \frac{1}{32768}\right) = \frac{32767}{512}$ ✓
 (≈ 63.998)

Approx decimal earns one mark only.

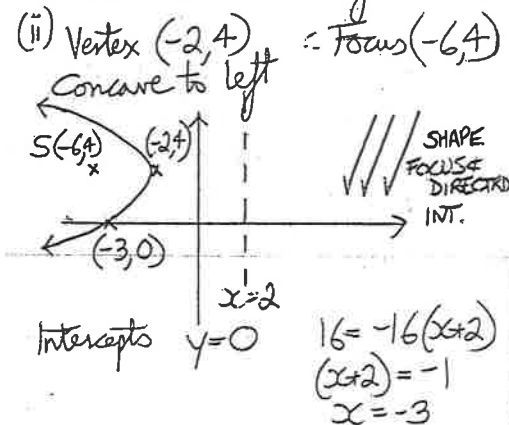
c) $\int_0^2 |x-1| dx = 0$ ✓
 (since triangles of equal area but one above & one below x-axis)

d) $\frac{2x-6}{x} \leq 1$ Domain $x \neq 0$
 (x^2)
 $(2x-6)x \leq x^2$ ✓
 $2x^2 - 6x - x^2 \leq 0$
 $x^2 - 6x \leq 0$
 $x(x-6) \leq 0$



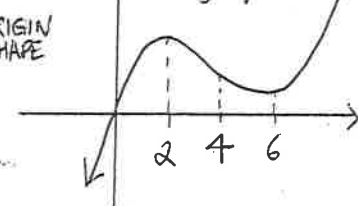
hence $0 \leq x \leq 6$ ✓✓

e) i) $4a = 16$
 $a = 4$. Focal length is $4a$ ✓



Question 6

a) Statpts @ $x=2$ and $x=6$
 $\checkmark$ $x=2$ Max t.p.
 $\checkmark$ $x=6$ Min t.p.
 $\checkmark$ $x=4$ Point of inflection
 $\checkmark$ ORIGIN & SHAPE



b) i)

x	0	2	4	6	8
$f(x)$	$\ln 1$	$\ln 5$	$\ln 7$	$\ln 37$	$\ln 65$

$\int_0^8 f(x) dx \approx \frac{1}{2} \times 4 [\ln 1 + \ln 7] + \frac{1}{2} \times 4 [\ln 7 + \ln 65]$ ✓
 $\approx 2 [2\ln 7 + \ln 65]$ ✓
 ≈ 19.7 (1dp) ✓

(ii) $\frac{1}{6} \times 4 \times (0 + 4\ln 5 + \ln 17) + \frac{1}{6} \times 4 (\ln 7 + 4\ln 37 + \ln 65)$ ✓
 ≈ 20.48 (3dp) ✓

c) i) $y = \frac{\ln x}{x^2}$
 $\frac{dy}{dx} = \frac{x^2(\frac{1}{x}) - \ln x(2x)}{x^4}$ ✓
 $= \frac{x - 2x \ln x}{x^4}$
 $= \frac{1 - 2 \ln x}{x^3}$ ✓

$$(ii) y = e^{2x}(2-x^2)$$

$$\frac{dy}{dx} = 2e^{2x}(2-x^2) + e^{2x}(-2x)$$

$$= 2e^{2x}[2-x-x^2]$$

$$= -2e^{2x}(x^2+x-2)$$

$$= -2e^{2x}(x+2)(x-1)$$

$$b) \int \frac{4x}{x^2+3} dx = 2 \int \frac{2x}{x^2+3} dx$$

$$= [2 \ln(x^2+3)]$$

$$= 2 \ln 3 - 2 \ln 4$$

$$= 2 \ln \frac{3}{4} \text{ or } -2 \ln \frac{4}{3}$$

$$e) 9^x - 10 \times 3^x + 9 = 0$$

$$(3^x)^2 - 10 \times 3^x + 9 = 0$$

$$(3^x - 9)(3^x - 1) = 0$$

$$3^x = 9 \text{ or } 3^x = 1$$

$$\therefore x = 2 \text{ or } x = 0$$

Question 7

$$a) 2x^2 + 4x - 1 = 0$$

$$a=2 \quad b=4 \quad c=-1$$

$$(i) \alpha\beta = \frac{c}{a} = \left(-\frac{1}{2}\right)$$

$$(ii) \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta}$$

$$= \frac{-b/a}{c/a}$$

$$= \frac{-b}{c}$$

$$= 4$$

$$(ii) (\alpha-\beta)^2 = \alpha^2 - 2\alpha\beta + \beta^2$$

$$= \alpha^2 + 2\alpha\beta + \beta^2 - 4\alpha\beta$$

$$= (\alpha+\beta)^2 - 4\alpha\beta$$

$$= \left(-\frac{1}{2}\right)^2 - 4 \times \left(-\frac{1}{2}\right)$$

$$= \frac{1}{4} + 2 = \frac{9}{4}$$

$$\therefore \alpha-\beta = \pm \frac{3}{2}$$

$$(i) |\alpha-\beta| = \sqrt{(\alpha-\beta)^2}$$

$$= \sqrt{\frac{9}{4}} = \frac{3}{2}$$

$$b) (i) AP \quad a=105 \quad d=7$$

$$t_n = a + (n-1)d$$

$$= 105 + 7(n-1)$$

$$t_n < 1000$$

$$105 + 7(n-1) < 1000$$

$$98 + 7n < 1000$$

$$7n < 902$$

$$n < \frac{902}{7}$$

$$n < 128\frac{4}{7}$$

$$t_{128} = 105 + 7 \times 127$$

$$= 105 + 889$$

$$= 994$$

$$\therefore 128 \text{ multiples of } 7 \text{ between } 100 \text{ and } 1000$$

$$(ii) S_{128} = \frac{n}{2}(a+l)$$

$$= \frac{128}{2}(105 + 994)$$

$$= 70336$$

$$c) (i) 2x^2 - 4x - 6 = 8x - 16$$

$$2x^2 - 12x + 10 = 0$$

$$x^2 - 6x + 5 = 0$$

$$(x-1)(x-5) = 0$$

$$x=1 \text{ or } x=5$$

$$(ii) \text{Area} = \int_1^5 (8x-16) - (2x^2-4x-6) dx$$

$$= \int_1^5 (12x - 10 - 2x^2) dx$$

$$= \left[6x^2 - 10x - \frac{2x^3}{3}\right]_1^5$$

$$= \left(150 - 50 - \frac{250}{3}\right) - \left(6 - 10 - \frac{2}{3}\right)$$

$$= 104 - \frac{248}{3}$$

$$= 104 - 82\frac{2}{3}$$

$$= 21\frac{2}{3}$$

$$d) (i) y = mx$$

$$(x+3)^2 + y^2 = 1$$

$$\text{Sub ① into ②}$$

$$x^2 + 6x + 9 + m^2 x^2 = 1$$

$$(1+m^2)x^2 + 6x + 8 = 0$$

$$\Delta = 36 - 4 \times (1+m^2) \times 8$$

$$= 36 - 32(1+m^2)$$

$$= 4 - 32m^2$$

$$\text{For tangency } \Delta = 0$$

$$m^2 = \frac{1}{8}$$

$$m = \pm \frac{1}{2\sqrt{2}}$$

$$\therefore \text{Eqns of tangents}$$

$$y = \frac{x}{4} \text{ or } y = -\frac{\sqrt{2}x}{4}$$

Question 8

$$a) (i) y = \frac{1}{2} \ln(3-x) \quad ①$$

$$V_y = \pi \int x^2 dy$$

$$\text{from ① } 2y = \ln(3-x)$$

$$e^{2y} = (3-x)$$

$$x = 3 - e^{2y}$$

$$(ii) x^2 = 9 - 6e^{2y} + e^{4y}$$

$$V = \pi \int_{-\frac{1}{2}\ln 3}^{\frac{1}{2}\ln 3} (9 - 6e^{2y} + e^{4y}) dy$$

$$= \pi \left[9y - \frac{6e^{2y}}{2} + \frac{e^{4y}}{4}\right]_{-\frac{1}{2}\ln 3}^{\frac{1}{2}\ln 3}$$

$$= \pi \left[\frac{9}{2} \ln 3 - 9 + \frac{9}{4} - \left(0 - 3 + \frac{1}{4}\right)\right]$$

$$= \pi \left(\frac{9}{2} \ln 3 - 4\right) u^3$$

$$b) y = (x-2)e^{-x}$$

$$(i) y \text{ into } x=0$$

$$y = -2 \times e^{-0} = -2 \quad (0, -2)$$

$$x \text{ into } y=0$$

$$e^{-x} \neq 0 \therefore x-2=0$$

$$x=2$$

$$(2, 0)$$

$$(ii) \frac{dy}{dx} = -e^{-x}(x-2) + 1 \times e^{-x}$$

$$= e^{-x}[3-x]$$

$$\text{Stat pts } \frac{dy}{dx} = 0$$

$$e^{-x} \neq 0 \therefore x=3 \quad (3, e^3)$$

$$y = \frac{1}{e^3}$$

$$\frac{d^2y}{dx^2} = -e^{-x}(3-x) + e^{-x}(-1)$$

$$= e^{-x}[x-4]$$

$$\therefore \left(\frac{d^2y}{dx^2}\right)_{x=3} = -\frac{1}{e^3} < 0 \therefore \text{Max}$$

$$\text{pt at } (3, \frac{1}{e^3})$$

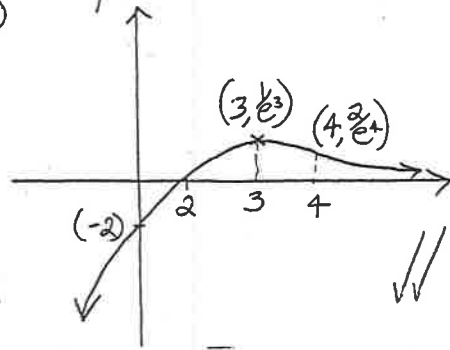
(iii) Possible POF $\frac{dy}{dx} = e^{-x}(x-4)$
 $\frac{dy}{dx} = 0 \quad e^{-x} \neq 0 \therefore x=4$
 $y = \frac{2}{e^4}$

x	3	4	5
$\frac{dy}{dx}$	$-\frac{1}{e^3}$	0	$\frac{1}{e^5}$

There is a change in concavity hence $(4, \frac{2}{e^4})$ is a point of inflection

(iv) $y = (x-2)e^{-x}$
 as $x \rightarrow \infty \quad xe^{-x} \rightarrow 0^+$
 $\therefore (x \text{ axis}) y=0$ is an asymptote

(v)



Question 9

a) i) Area $\triangle ACD = \frac{1}{2}bc \sin(A\hat{C}D)$
 or $= \frac{1}{2}bc \sin(\beta - \alpha)$
 Area $\triangle ABD = \frac{1}{2}ab \sin \alpha$
 Area $\triangle ABC = \frac{1}{2}ac \sin \beta$
 $\therefore$ Area $\triangle ACD = \frac{1}{2}ac \sin \beta - \frac{1}{2}ab \sin \alpha$

Equating area expressions
 $\frac{1}{2}bc \sin(\beta - \alpha) = \frac{1}{2}ab \sin \beta - \frac{1}{2}ac \sin \alpha$
 in $\triangle ABD$
 $\cos \alpha = \frac{a}{b}$
 $\therefore a = b \cos \alpha$ ②
 in $\triangle ABC$
 $\cos \beta = \frac{a}{c}$
 $\therefore a = c \cos \beta$ ③
 Sub ② and ③ into ①
 $\frac{1}{2}bc \sin(\beta - \alpha) = \frac{1}{2}(b \cos \alpha) \sin \beta - \frac{1}{2}(c \cos \beta) \sin \alpha$

($\div \frac{1}{2}bc$)
 $\sin(\beta - \alpha) = \sin \beta \cos \alpha - \sin \alpha \cos \beta$

b) $y = \sqrt{x} e^{x^2}$
 $= x^{\frac{1}{2}} e^{x^2}$
 (i) $\frac{dy}{dx} = \frac{1}{2}x^{-\frac{1}{2}} e^{x^2} + x^{\frac{1}{2}} \times 2x e^{x^2}$
 $= \frac{e^{x^2}}{2\sqrt{x}} + \frac{e^{x^2}}{2}$

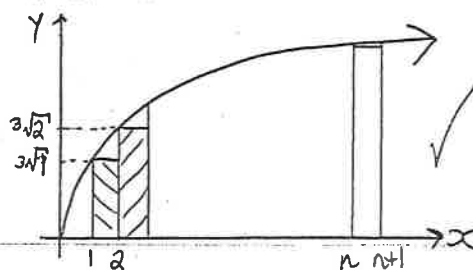
(ii) $\int e^{x^2} dx$ Hence integrate result
 $\sqrt{x} e^{x^2} + c = \int \frac{e^{x^2}}{2\sqrt{x}} dx + \int \frac{e^{x^2}}{2} dx$

$\sqrt{x} e^{x^2} + c = e^{x^2} + \frac{1}{2} \int e^{x^2} dx$
 $\therefore \frac{1}{2} \int e^{x^2} dx = e^{x^2}(\sqrt{x} - 1) + c$
 $\int e^{x^2} dx = 2e^{x^2}(\sqrt{x} - 1) + d$

(iii) $\int_0^4 e^{x^2} dx = [2e^{x^2}(\sqrt{x} - 1)]_0^4$
 $= 2e^4 + 2 = 2(e^4 + 1)$

c) Area under cube root curve is overestimated by upper rectangles since $\sqrt[3]{x}$ is concave down.
 Height of each rectangle is at coord at right hand edge cube rooted, width of each rectangle is one
 ie first rectangle $A_1 = 1 \times \sqrt[3]{1}$
 last rectangle $A_n = 1 \times \sqrt[3]{n}$
 hence $\sqrt[3]{1} + \sqrt[3]{2} + \dots + \sqrt[3]{n} > \int_0^n \sqrt[3]{x} dx$

(ii) This set of rectangles must underestimate area under curve hence take height as cube root of x coordinate at left hand end of rectangle to get lower rectangles as illustrated below



hence $\int_1^{n+1} \sqrt[3]{x} dx > \sqrt[3]{1} + \sqrt[3]{2} + \dots + \sqrt[3]{n}$
 note change to limits of integral $1 \rightarrow n+1$

(iv) $\int_0^{100} \sqrt[3]{x} dx < \sqrt[3]{1} + \sqrt[3]{2} + \sqrt[3]{3} + \dots + \sqrt[3]{100} < \int_0^{101} \sqrt[3]{x} dx$
 $\left[\frac{3x^{4/3}}{4} \right]_0^{100} < \sum_{n=1}^{100} \sqrt[3]{n} < \left[\frac{3x^{4/3}}{4} \right]_0^{101}$
 $\frac{3}{4}(100^{4/3}) < \sum_{n=1}^{100} \sqrt[3]{n} < \frac{3}{4}(101^{4/3})$
 $348.1 \dots < \sum_{n=1}^{100} \sqrt[3]{n} < 352.0 \dots$
 $\sum_{n=1}^{100} \sqrt[3]{n} \approx 350$ (2sf)

(Must consider both bounds for two marks)

d) $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + \dots$
 $xe^x = x + \frac{x^2}{1!} + \frac{x^3}{2!} + \frac{x^4}{3!} + \dots + \frac{x^{n+1}}{n!} + \dots$
 $\frac{d}{dx}(xe^x) = 1 + 2x + \frac{3x^2}{2!} + \frac{4x^3}{3!} + \dots + \frac{(n+1)x^n}{n!} + \dots$
 LHS $= xe^x + 1 \times e^x = e^x(x+1)$
 hence required result is $e^x(x+1)$